ROBO SIDDHI

AI VISION

From Learning AI to Building AI

INTRODUCTION

"At RoboSiddhi, students learn Artificial Intelligence by building real, working projects. Instead of learning AI only through theory, students understand the complete journey — from identifying a problem to designing an idea, writing code, integrating AI models, testing the system and improving it."

WHY THESE PROJECTS?

These projects are designed to turn AI concepts into practical experiences. Students work with real-world problems such as attendance automation, computer vision, emotion recognition, object detection, driver safety and generative AI.

THE AI PROJECT LAB

PROJECT 00

STEM ATTENDANCE AI

WHAT IT DOES A face-recognition attendance system that identifies students through a camera and automatically records attendance.

HOW BUILT Built with computer vision, face recognition and a browser-based web application.

STUDENTS LEARN Face Recognition • Computer Vision • Python • Web Applications • Automation

REAL-WORLD SCOPE Smart attendance, contactless check-in and identity-based automation.

PROJECT 01

STEM EMOTION

WHAT IT DOES Reads facial expressions through a camera and identifies emotions such as neutral, happy and surprised in real time.

HOW BUILT Uses a trained vision model with real-time camera inference.

STUDENTS LEARN Computer Vision • Facial Features • AI Models • Real-Time Inference

REAL-WORLD SCOPE Human-computer interaction, interactive applications and vision-based systems.

PROJECT 02

STEM DETECTION

WHAT IT DOES Uses a camera to identify whether the visible subject is a cat, dog or human.

HOW BUILT A browser-based AI model performs real-time image classification and detection.

STUDENTS LEARN Object Detection • Machine Learning • Camera Processing • Model Inference

REAL-WORLD SCOPE Smart cameras, object recognition and automated monitoring.

PROJECT 03

STEM DRIVER

WHAT IT DOES Monitors driver-related visual signals such as yawning, drowsiness and phone use.

HOW BUILT Uses MediaPipe and COCO-SSD with real-time, on-device computer vision.

STUDENTS LEARN Computer Vision • MediaPipe • Object Detection • Real-Time AI

REAL-WORLD SCOPE Driver monitoring, safety research and intelligent assistance systems.

PROJECT 04

EMPIRE AI

WHAT IT DOES An all-in-one AI workspace combining AI chat, web research, document intelligence, coding, RAG and AI agents.

HOW BUILT Combines modern Generative AI, multi-model interaction, retrieval-based workflows and AI agent concepts.

STUDENTS LEARN Generative AI • LLMs • RAG • AI Agents • Prompt Engineering

REAL-WORLD SCOPE AI assistants, research automation, document intelligence and productivity tools.

HOW EVERY PROJECT WORKS

THE AI DEVELOPMENT CYCLE

1. PROBLEM
2. IDEA
3. CODE
4. AI MODEL
5. TEST
6. IMPROVE

Students experience the complete AI development cycle through practical projects.

ROBO SIDDHI

AI + VISION + FUTURE SKILLS

Real Projects. Real AI. Real Learning.

THE AI PROJECT LAB

PROJECT 05

STEM CONTROLLER

WHAT IT DOES A Snake game controlled through keyboard input or by tracking the user's index finger through a webcam.

HOW BUILT Uses hand tracking and computer vision to convert finger movement into game controls.

STUDENTS LEARN Hand Tracking • Gesture Recognition • Computer Vision • Real-Time Processing

REAL-WORLD SCOPE Gesture-controlled interfaces, gaming and touchless interaction.

PROJECT 06

STEM VISION LAB

WHAT IT DOES An interactive computer vision laboratory with real-time AI and machine learning experiments.

HOW BUILT Uses browser-based computer vision and machine learning for interactive experimentation.

STUDENTS LEARN Image Processing • Machine Learning • Computer Vision • Real-Time AI

REAL-WORLD SCOPE AI experimentation, rapid prototyping and computer vision applications.

PROJECT 07

CALCULATIVE TUG OF WAR

WHAT IT DOES An interactive calculation tug-of-war game where players solve challenges and use their answers to power a win.

HOW BUILT Built to make maths engaging through play — answer challenges correctly and your side pulls harder with mental math speed.

STUDENTS LEARN Mathematics • Quick Math • Strategy • STEM Play

REAL-WORLD SCOPE Gamified math practice and classroom competitions with STEM learning through play.

PROJECT 08

STEM FACE MASK

WHAT IT DOES Detects whether a person is wearing a mask or not through a webcam in real time.

HOW BUILT Uses computer vision and browser-based model inference without sending camera frames to a server.

STUDENTS LEARN Image Classification • Computer Vision • Model Inference • Real-Time Detection

REAL-WORLD SCOPE Vision-based monitoring and classification systems.

PROJECT 09

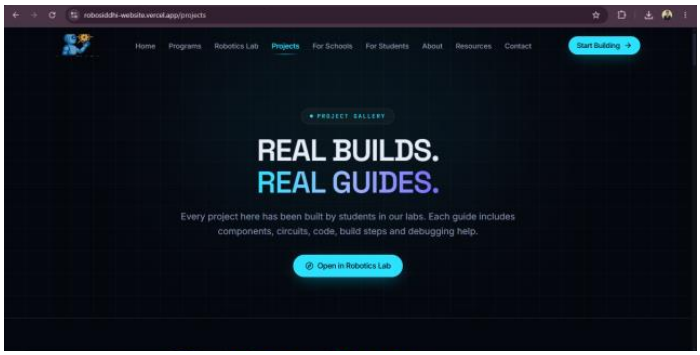
STEM SIGN LANGUAGE

WHAT IT DOES Recognizes hand gestures through a webcam and translates American Sign Language gestures into letters and words.

HOW BUILT Uses hand tracking, gesture recognition and browser-based AI inference.

STUDENTS LEARN MediaPipe Hands • Gesture AI • Pattern Recognition • TensorFlow.js

REAL-WORLD SCOPE Accessible technology, human-computer interaction and gesture-based communication.



SKILLS STUDENTS DEVELOP

▪ **COMPUTER VISION**	Understanding how machines interpret images, faces, gestures and objects.
▪ **AI & MACHINE LEARNING**	Understanding models, inference, predictions and practical AI applications.
▪ **PROGRAMMING**	Writing real code, connecting components and building working applications.
▪ **PROBLEM SOLVING**	Breaking real-world problems into logical and programmable solutions.
▪ **INNOVATION**	Turning ideas into prototypes and improving them through testing.

HOW ROBO SIDDHI TEACHES AI

LEARN → BUILD → TEST → IMPROVE → INNOVATE

AI USER → AI LEARNER → AI BUILDER

Our goal is not only to teach students how to use AI tools. We help students understand how AI systems work, build their own projects and develop the confidence to create technology.

EXPLORE THE LIVE PROJECTS

Instruction: Visit the RoboSiddhi website > Click Projects > Explore the AI Project Lab

robosiddhi-website.vercel.app/projects

ROBO SIDDHI

BUILD. CODE. CREATE. INNOVATE.

AI • Robotics • IoT • STEM | Jaipur, Rajasthan, India

Website: robosiddhi-website.vercel.app